MODULE 4 — 3-Mark Questions with Answers

A. Short Answer Questions (1–20)

1. Define SNMP. Name its two main components and state one function of each.

SNMP (Simple Network Management Protocol) is an application-layer protocol used to monitor and manage network devices such as routers, switches, and servers.

Its two main components are:

- Manager: Monitors and controls network devices by sending requests and receiving responses.
- Agent: Software running on a managed device that collects device information and responds to the manager.

2. What is a MIB? How is it different from SMI?

- MIB (Management Information Base): A database containing information about the managed objects of a network device.
- SMI (Structure of Management Information): Defines the rules and data types used to describe and identify objects in a MIB.

Difference: MIB stores the definitions of manageable objects, whereas SMI specifies how those objects are structured and represented.

3. List the symbols used in ASN.1 notation and give the meaning of any three.

ASN.1 (Abstract Syntax Notation One) uses symbols such as:

- `::=` — means "is defined as."
- `{ }` — enclose a list of components or values.
- `( )` — enclose constraints or numeric values.
- `|` — separates alternatives in some ASN.1 constructs.
- `,` — separates items in a list.

Any three: `::=`, `{ }`, and `( )` are commonly used in ASN.1 definitions.

4. Differentiate between analog data and digital data with one example each.

Analog data	Digital data
Varies continuously over time.	Represented using discrete values.
Can take infinitely many values within a range.	Commonly represented using binary 0s and 1s.

Example: Human voice waveform.	Example: Text stored in a computer.

5. Define bandwidth of a signal. Calculate its bandwidth if its frequency ranges from 1000 Hz to 5000 Hz.

Bandwidth is the difference between the highest and lowest frequencies present in a signal.

Formula:

B=fmax⁡−fmin⁡B=f_{\max}-f_{\min}

Given:

- Highest frequency = 5000 Hz
- Lowest frequency = 1000 Hz

Calculation:

B=5000−1000=4000 HzB=5000-1000=4000\text{ Hz}

Answer: Bandwidth = 4000 Hz or 4 kHz.

6. State the Nyquist sampling theorem. What is the Nyquist rate for a voice signal with maximum frequency 4000 Hz?

The Nyquist sampling theorem states that a signal must be sampled at a rate at least twice its highest frequency to reconstruct it accurately, assuming ideal band-limited sampling.

Formula:

fs≥2fmax⁡f_s\geq 2f_{\max}

Given:

fmax⁡=4000 Hzf_{\max}=4000\text{ Hz}

Therefore,

fs=2×4000=8000 samples/sf_s=2\times4000=8000\text{ samples/s}

Answer: Nyquist rate = 8000 samples per second (8 kHz).

7. Differentiate between bandwidth and throughput.

Bandwidth	Throughput
Maximum theoretical data-carrying capacity of a communication channel.	Actual amount of data successfully transmitted per unit time.

Indicates the potential capacity of a connection.	Indicates actual transmission performance.
Example: A 100 Mbps link.	Example: Actual data transfer of 80 Mbps.

8. Define attenuation. Write the formula used to express signal loss/gain in decibels.

Attenuation is the reduction in signal strength as it travels through a transmission medium.

The signal gain or loss in decibels (dB) is calculated using:

dB=10log⁡10(P2P1)\text{dB}=10\log_{10}\left(\frac{P_2}{P_1}\right)

Where:

- P1P_1 = input power
- P2P_2 = output power

A negative dB value indicates power loss, while a positive value indicates power gain.

9. What is jitter? Why is it a concern for real-time applications?

Jitter is the variation in the delay of packets as they travel through a network.

It is a concern for real-time applications because irregular packet arrival times can cause:

- Distorted audio or video.
- Interruptions during video calls.
- Uneven playback in streaming applications.

10. Differentiate between baseband and broadband transmission.

Baseband transmission	Broadband transmission
Uses the entire channel for a single digital signal at a time.	Divides the available frequency range into multiple channels.
Commonly uses digital signaling.	Commonly uses modulation to carry signals over separate frequency bands.
Example: Traditional Ethernet LAN transmission.	Example: Cable television transmission.

11. What is line coding? Name any three line coding schemes.

Line coding is the process of converting digital data bits into digital signal patterns suitable for transmission over a communication medium.

Three line coding schemes are:

1. NRZ (Non-Return-to-Zero)
2. Manchester encoding
3. AMI (Alternate Mark Inversion)

12. Define quantization error. State the maximum possible quantization error for a given step size Δ.

Quantization error is the difference between the original sampled amplitude and its quantized representation.

For a uniform quantizer using rounding to the nearest level, the maximum absolute quantization error is:

Emax⁡=Δ2E_{\max}=\frac{\Delta}{2}

Where Δ\Delta is the quantization step size.

Answer: Maximum quantization error = ±Δ/2.

13. Differentiate between ASK, FSK and PSK based on the parameter of the carrier that changes.

Modulation technique	Parameter changed
ASK (Amplitude Shift Keying)	Amplitude
FSK (Frequency Shift Keying)	Frequency
PSK (Phase Shift Keying)	Phase

All three techniques transmit digital data by varying a parameter of a carrier signal.

14. What is a constellation diagram? What do the X-axis and Y-axis represent?

A constellation diagram is a graphical representation of the possible symbols of a digital modulation scheme.

- X-axis: In-phase component (I).
- Y-axis: Quadrature component (Q).

Each point represents a possible transmitted symbol. The diagram helps analyze modulation schemes and noise-related errors.

15. Differentiate between FDM and TDM.

FDM	TDM
Frequency Division Multiplexing.	Time Division Multiplexing.
Divides the channel into separate frequency bands.	Divides channel access into separate time slots.

Multiple signals can be transmitted simultaneously on different frequencies.	Multiple signals share the channel at different times.
Example: Radio broadcasting.	Example: Digital telephony systems.

16. What is spread spectrum? Name its two main types.

Spread spectrum is a communication technique in which a signal is spread over a wider frequency bandwidth than the minimum required, using a spreading code or sequence.

Its two main types are:

1. FHSS (Frequency Hopping Spread Spectrum): The carrier frequency changes among different frequencies according to a predetermined sequence.
2. DSSS (Direct Sequence Spread Spectrum): The data is combined with a high-rate spreading code to spread the signal over a wider bandwidth.

17. List any three types of guided transmission media.

Guided transmission media carry signals through a physical path.

The three main types are:

1. Twisted-pair cable
2. Coaxial cable
3. Fiber-optic cable

18. Why is twisted-pair cable twisted? What problem does this solve?

The wires in a twisted-pair cable are twisted together to reduce electromagnetic interference and crosstalk.

Twisting helps external interference affect both wires similarly, allowing much of the unwanted noise to be cancelled at the receiver.

19. Differentiate between UTP and STP.

UTP	STP
Unshielded Twisted Pair.	Shielded Twisted Pair.
Does not have an additional metallic shield.	Includes shielding to reduce electromagnetic interference.
Cheaper, lighter, and easier to install.	Generally more expensive and requires proper grounding where applicable.
Suitable for ordinary LAN environments.	Useful in environments with higher electromagnetic interference.

20. State two advantages of fiber-optic cable over coaxial cable.

Two advantages of fiber-optic cable are:

1. Higher bandwidth: It can support very high data rates over long distances.
2. Immunity to electromagnetic interference: It transmits data using light, so it is not affected by electromagnetic interference in the same way as coaxial cable.

Other advantages include lower signal attenuation over many long-distance links, electrical isolation, and greater resistance to electrical surges.

Quick revision: Important formulas

Topic	Formula / result
Bandwidth	B=fmax−fminB=f_{\max}-f_{\min}
Nyquist sampling rate	fs=2fmaxf_s=2f_{\max}
Power ratio in dB	10log10(P2/P1)10\log_{10}(P_2/P_1)
Maximum quantization error	±Δ/2\pm\Delta/2
Voice signal sampling at 4 kHz maximum frequency	8,000 samples/s

Exam tip: For 3-mark questions, write the definition first, followed by two or three key points. For numerical questions, show the formula, substitution, and final answer with units.